FlowchartFun

Answer key

OCR GCSE Computer Science (J277) flowchart practice.

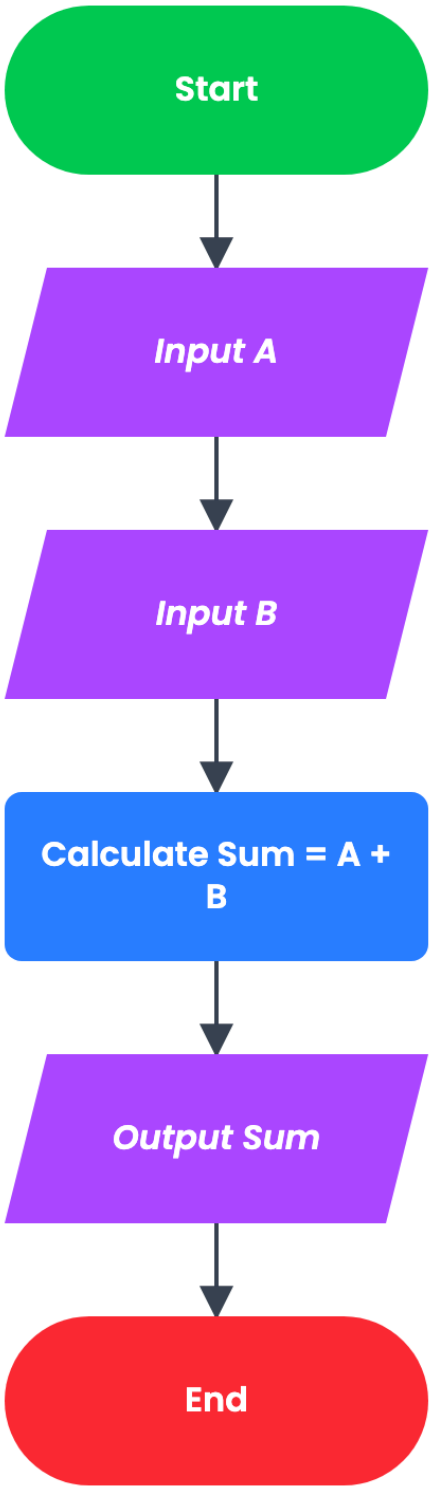
Completed flowcharts for every level. The same answers apply to all three modes.

Level 1: Flowchart Basics

Easy · Answer Key

Create a simple flowchart that calculates the sum of two numbers. Drag each symbol into the correct place.

Answer key

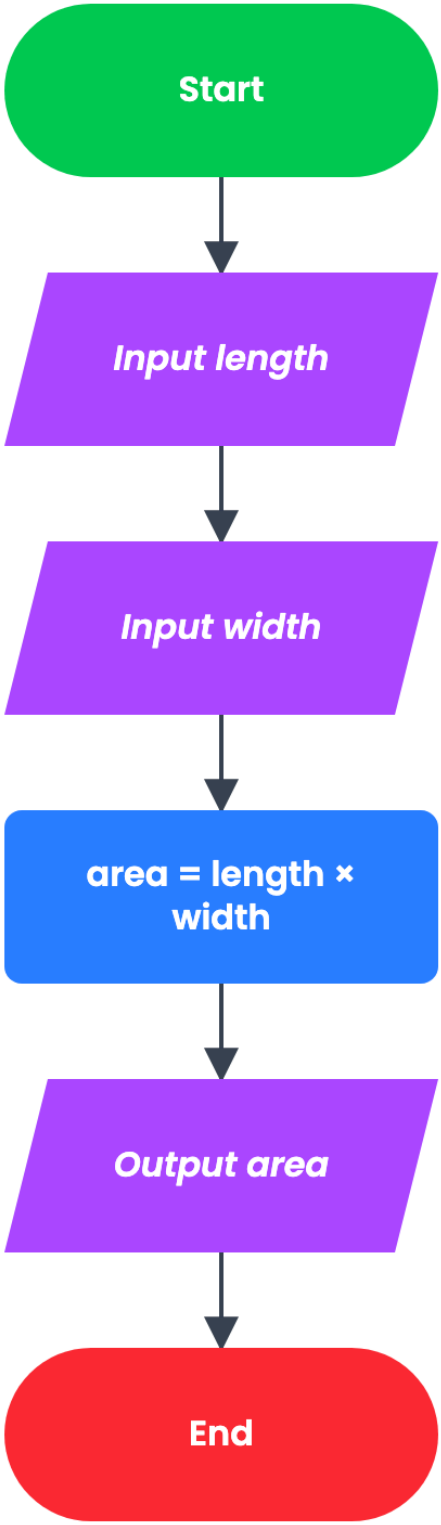


Level 2: Input, Process, Output

Easy · Answer Key

Make a flowchart that finds the area of a rectangle. Watch out: one symbol is not needed!

Answer key

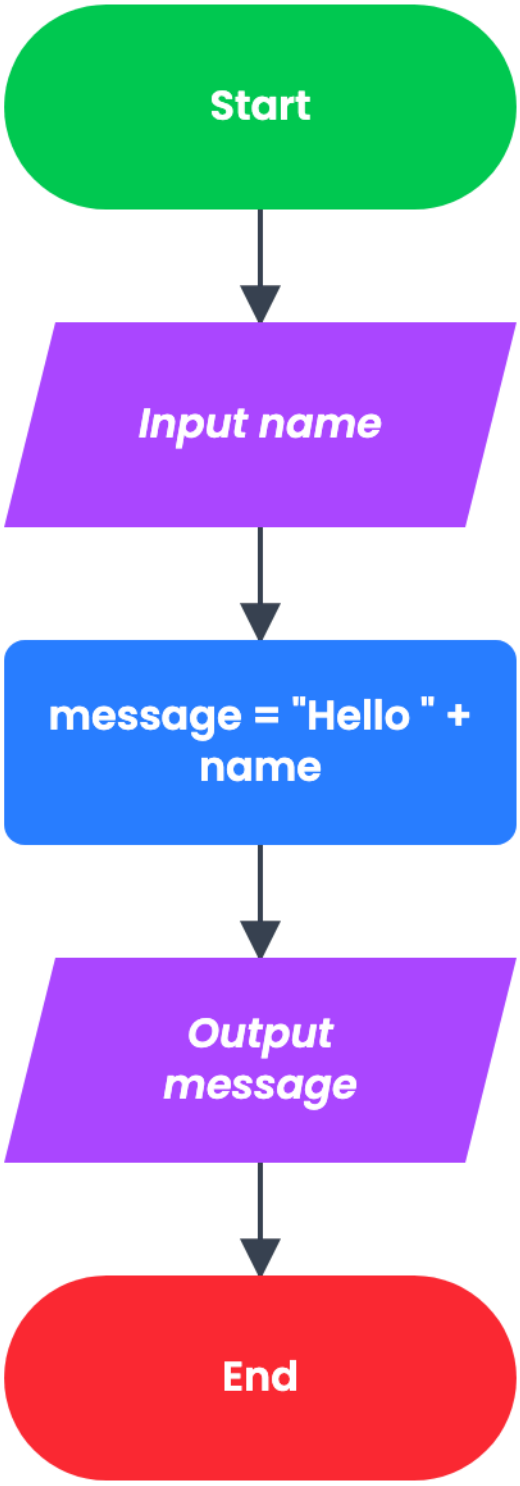


Level 3: Hello, Name

Easy · Answer Key

Ask for a name and greet the user. Not every symbol is needed.

Answer key

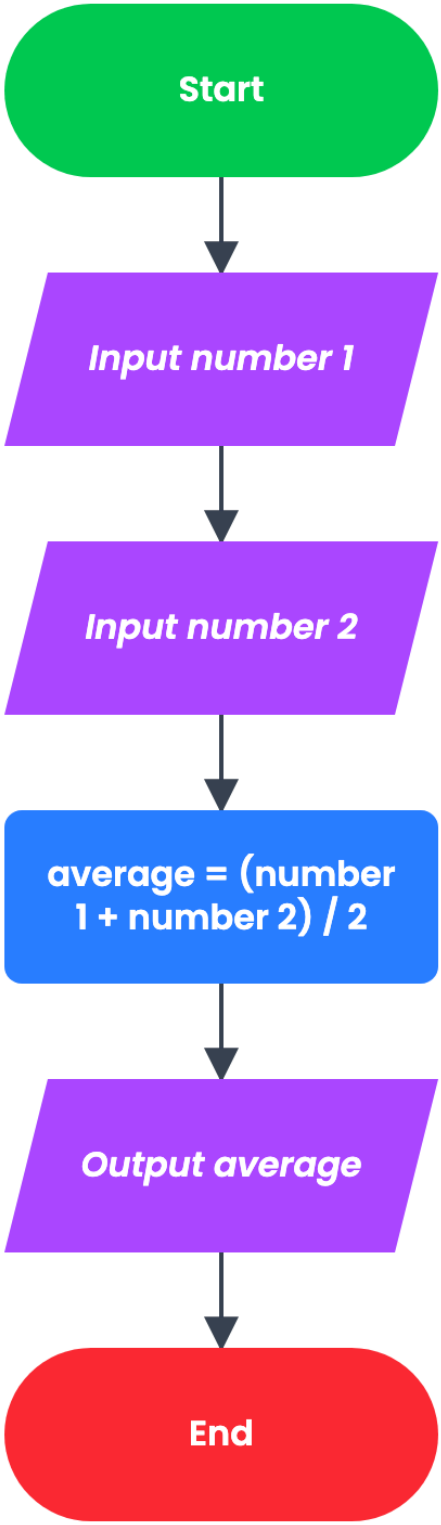


Level 4: Average of Two

Easy · Answer Key

Input two numbers and output their average. Remember the brackets!

Answer key

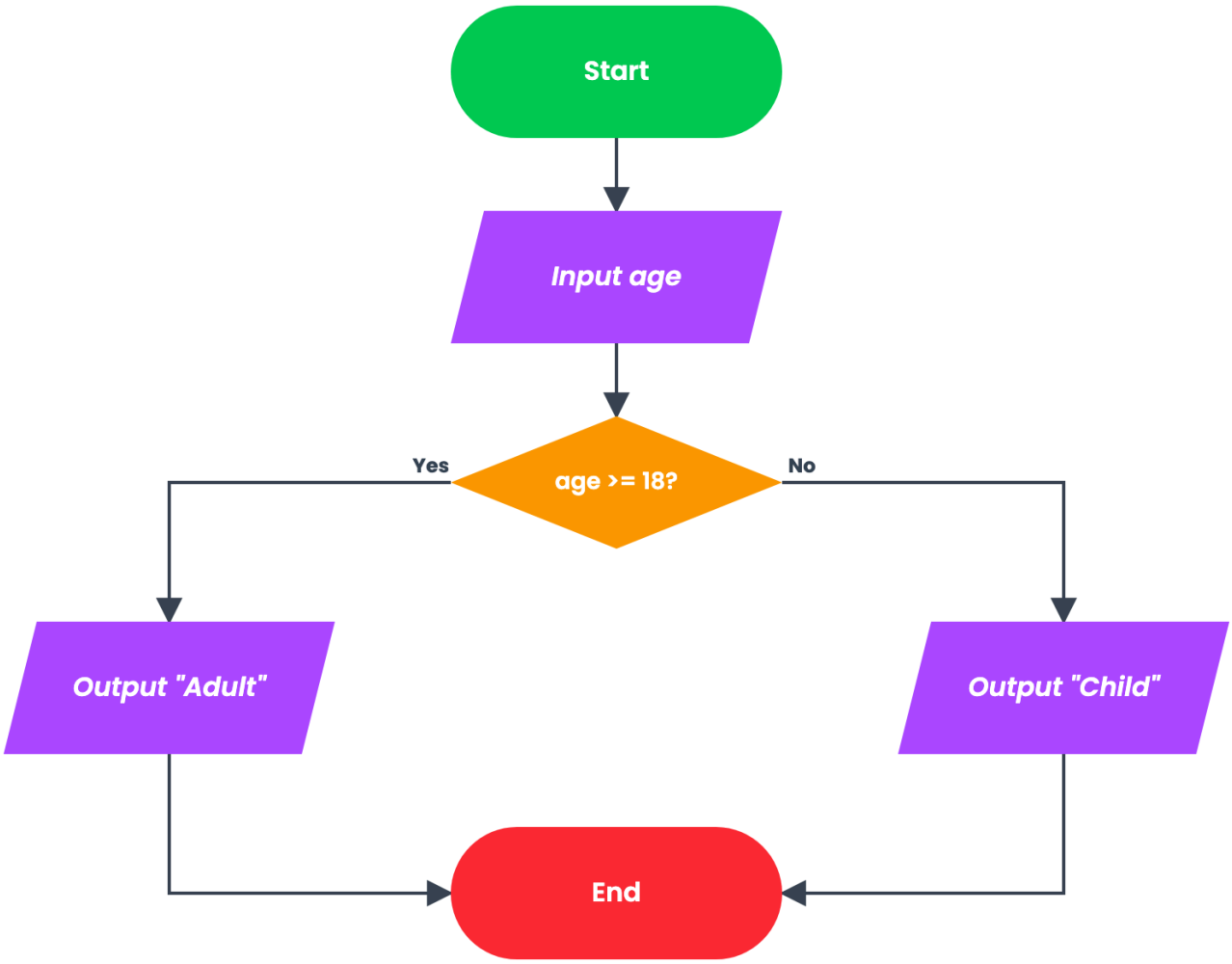


Level 5: Making a Decision

Easy · Answer Key

Output whether a person is an adult or a child. Remember: diamonds are decisions, with a Yes path and a No path.

Answer key

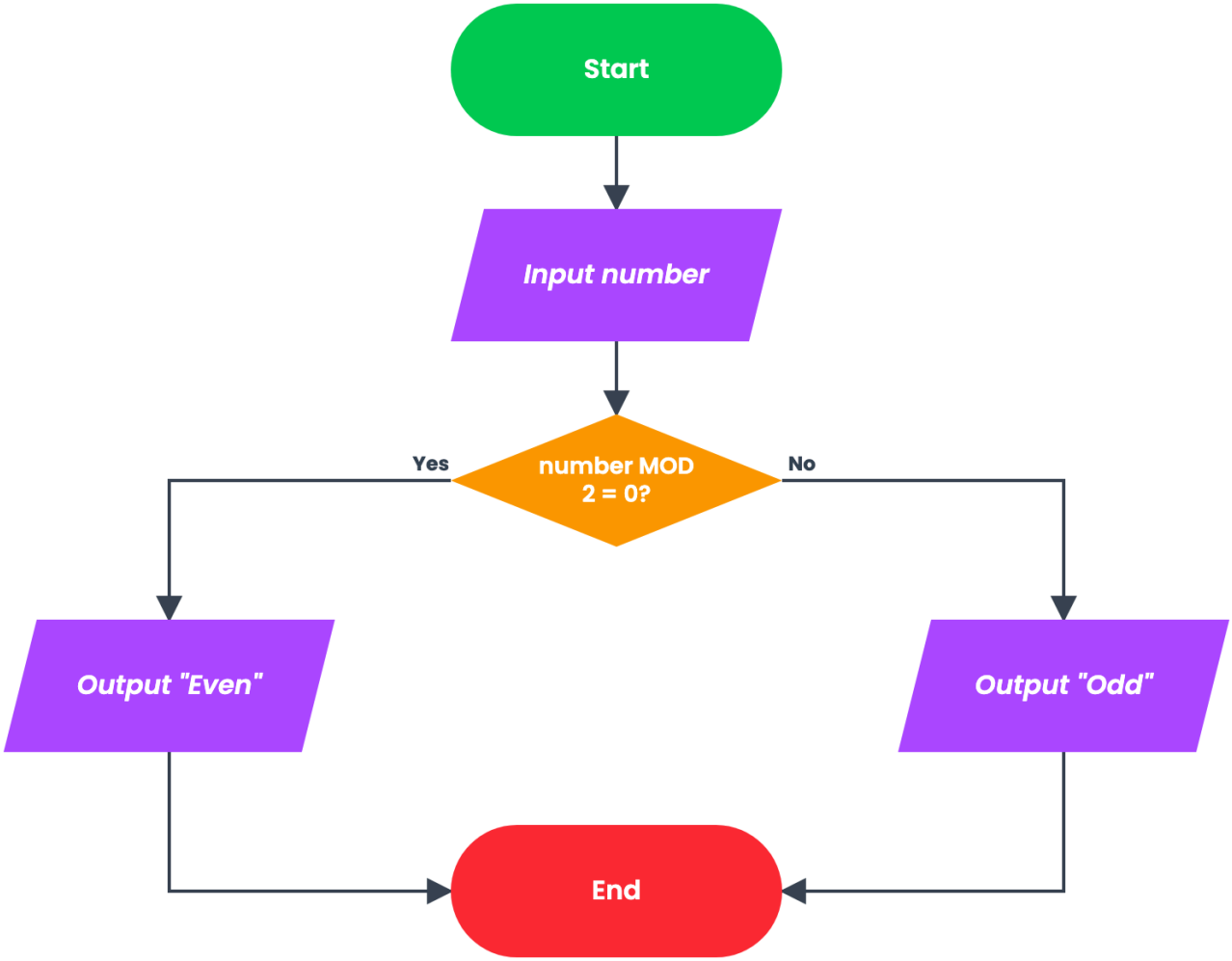


Level 6: Odd or Even

Easy · Answer Key

Use MOD (the remainder after division) to decide whether a number is even or odd.

Answer key

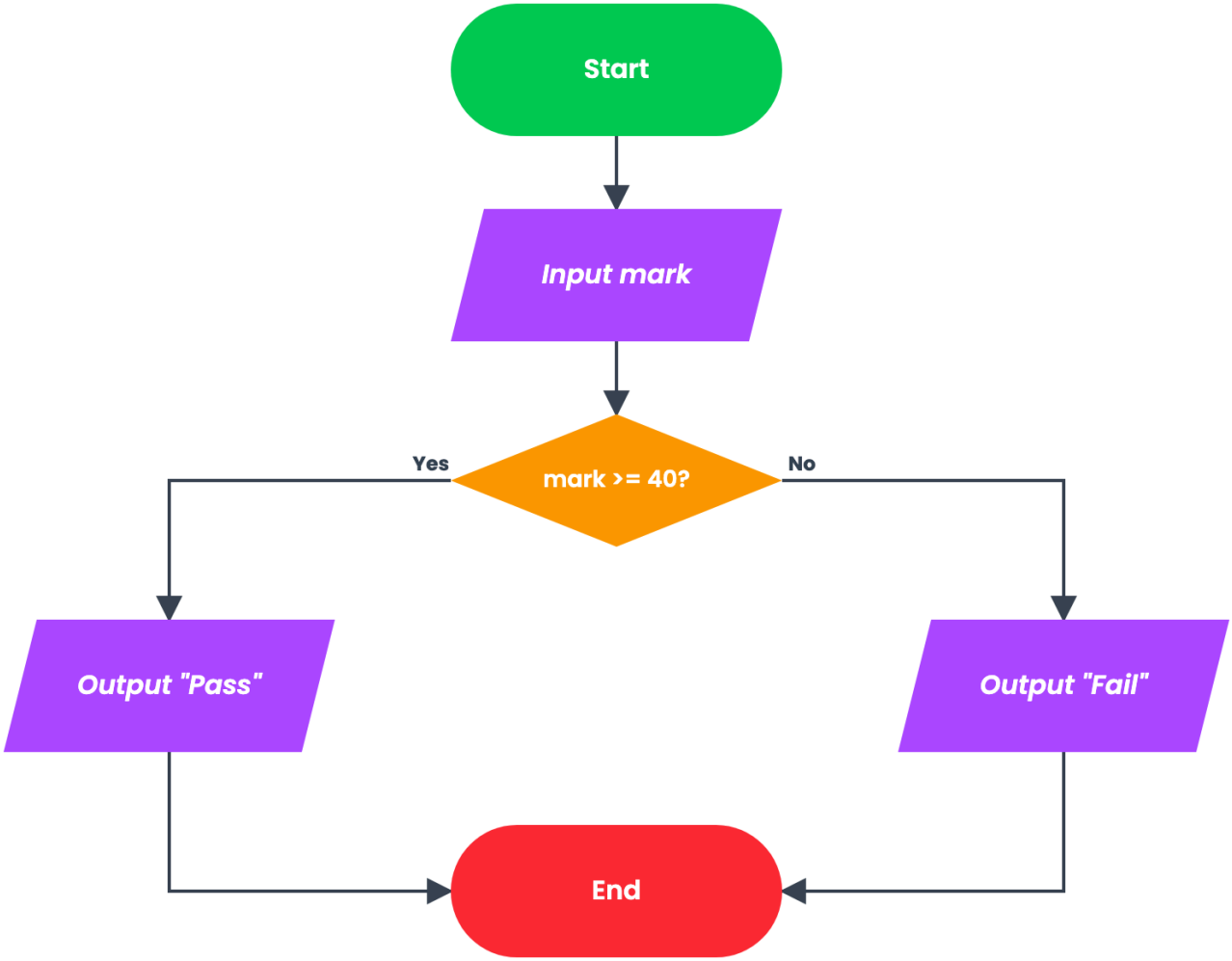


Level 7: Pass or Fail

Easy · Answer Key

A mark of 40 or more is a pass. Anything else is a fail.

Answer key

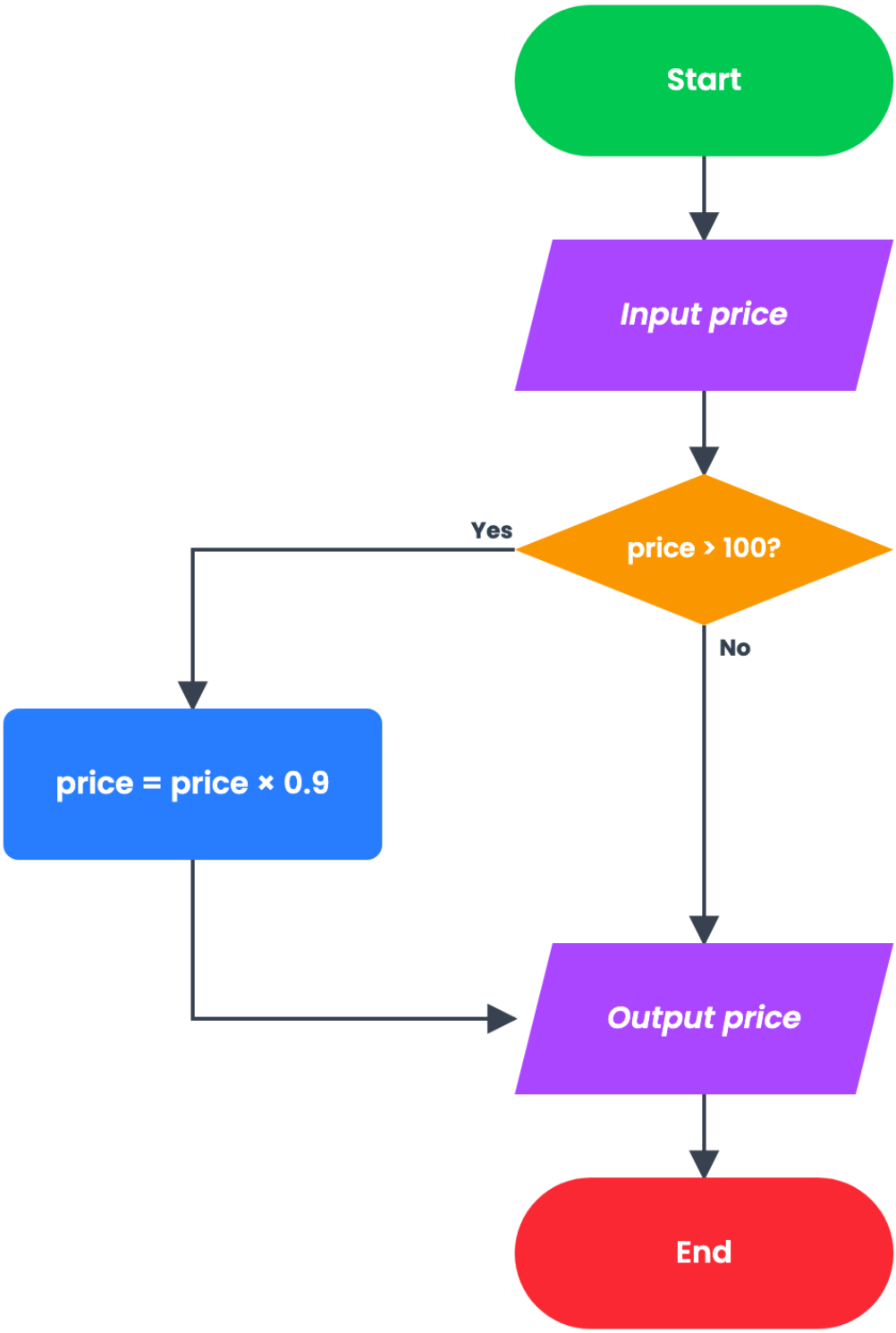


Level 8: Discount

Easy · Answer Key

Shoppers get 10% off if they spend more than £100. Everyone sees a price at the end. Notice that one path skips a step.

Answer key

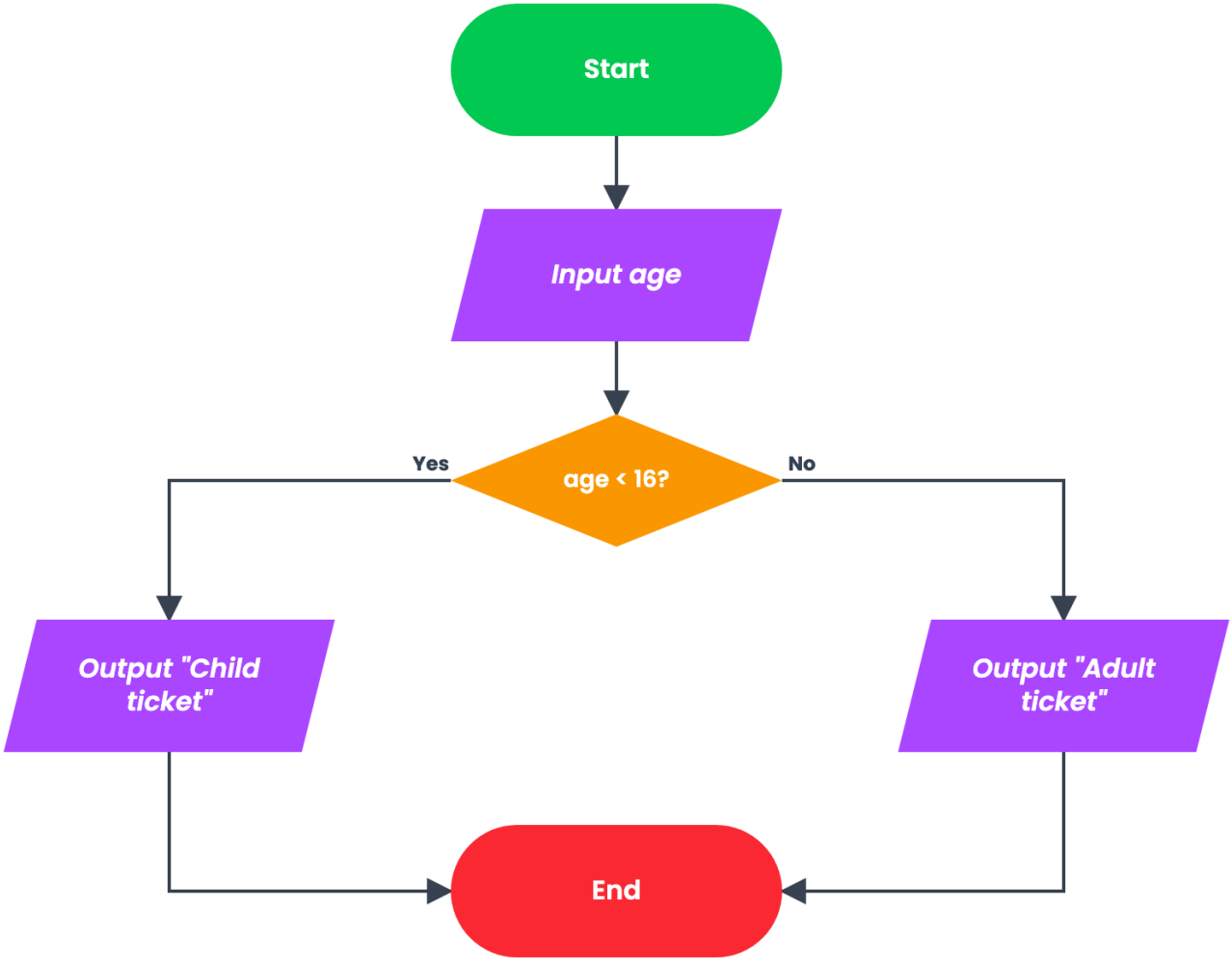


Level 9: Cinema Ticket

Easy · Answer Key

Under 16s pay the child price. Everyone else pays the adult price.

Answer key

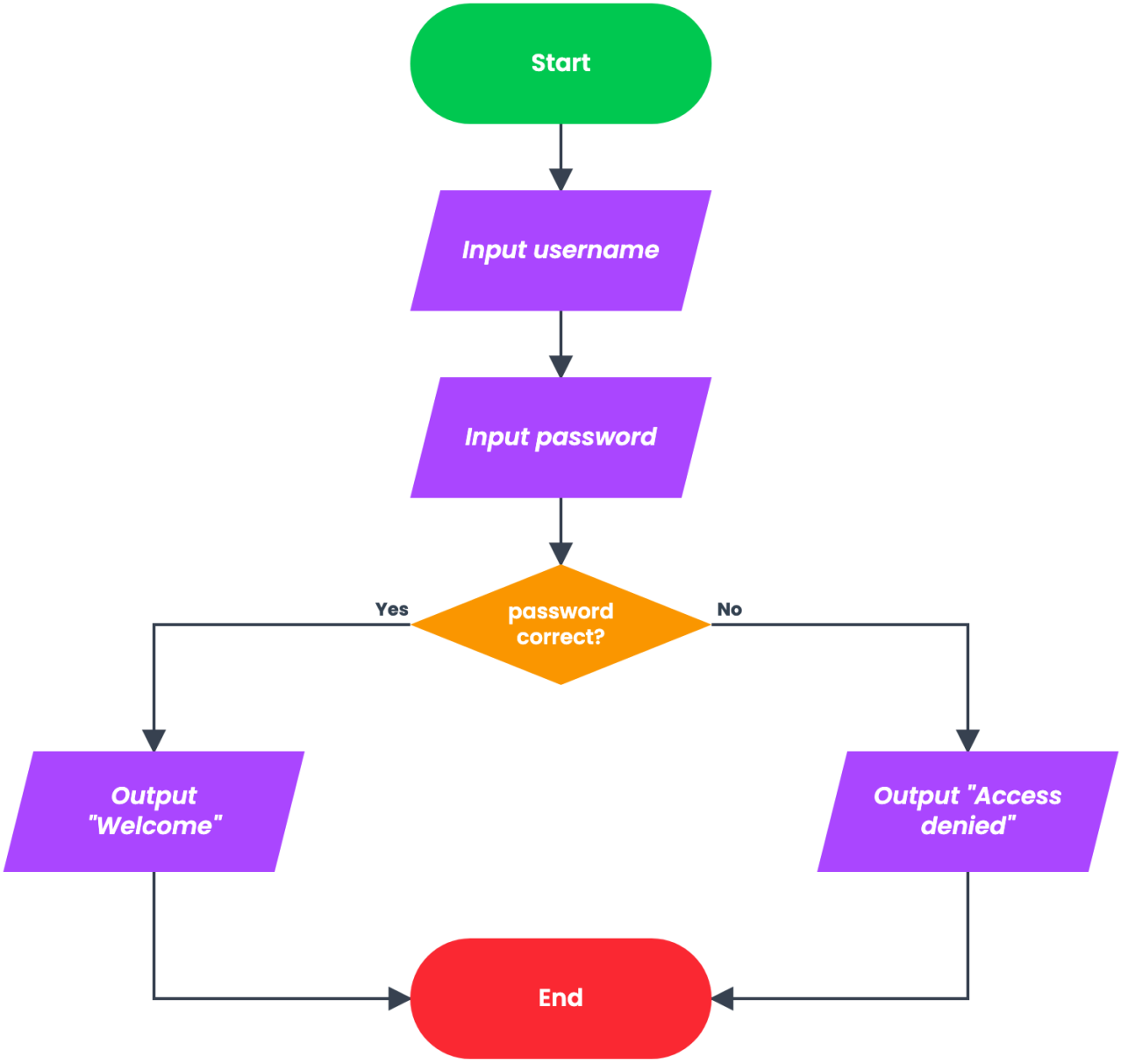


Level 10: Login

Easy · Answer Key

Ask for a username and password, then decide whether to let the user in.

Answer key

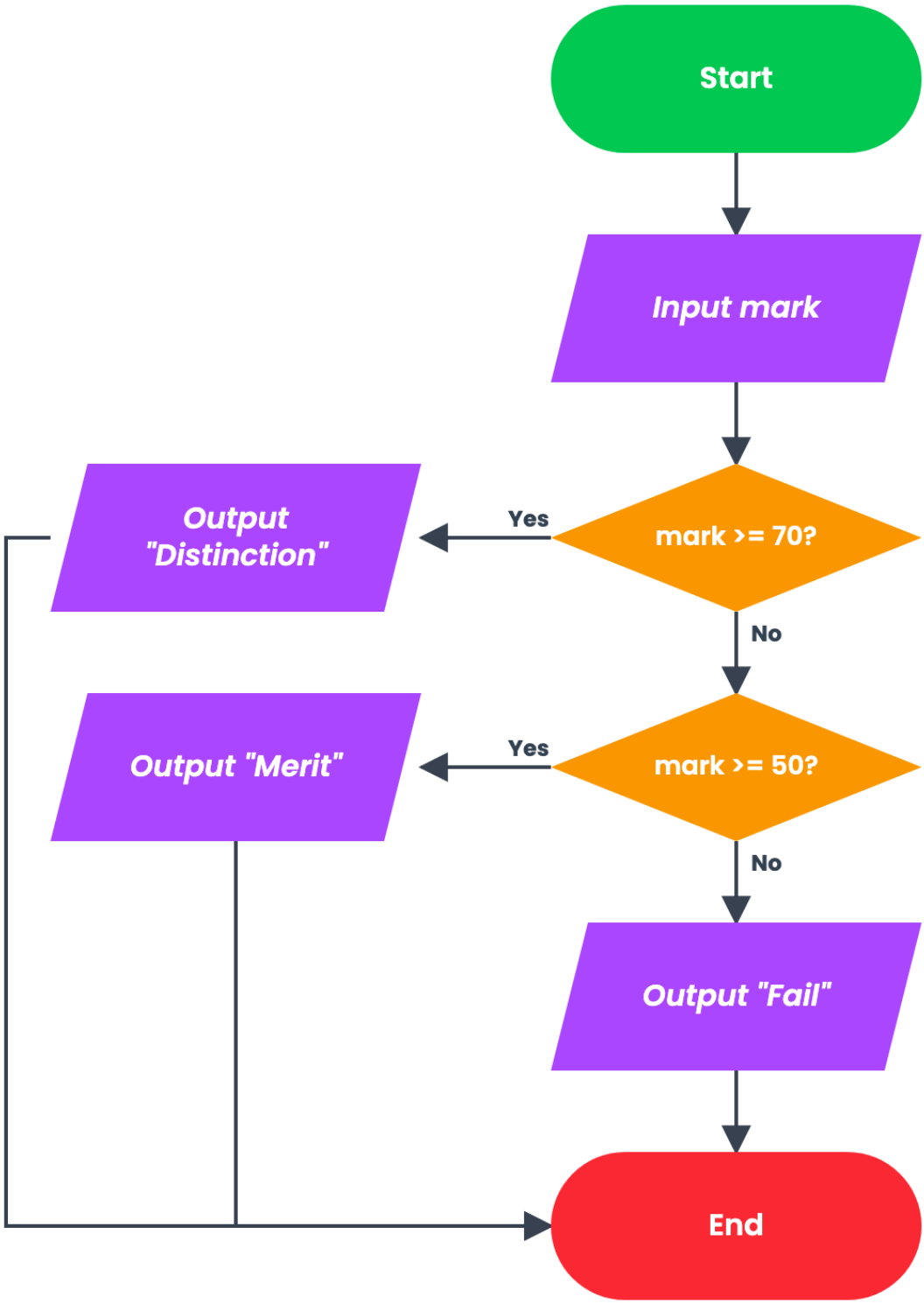


Level 11: Grade Boundaries

Medium · Answer Key

Output Distinction for 70 or more, Merit for 50 or more, otherwise Fail. Start is already placed for you.

Answer key

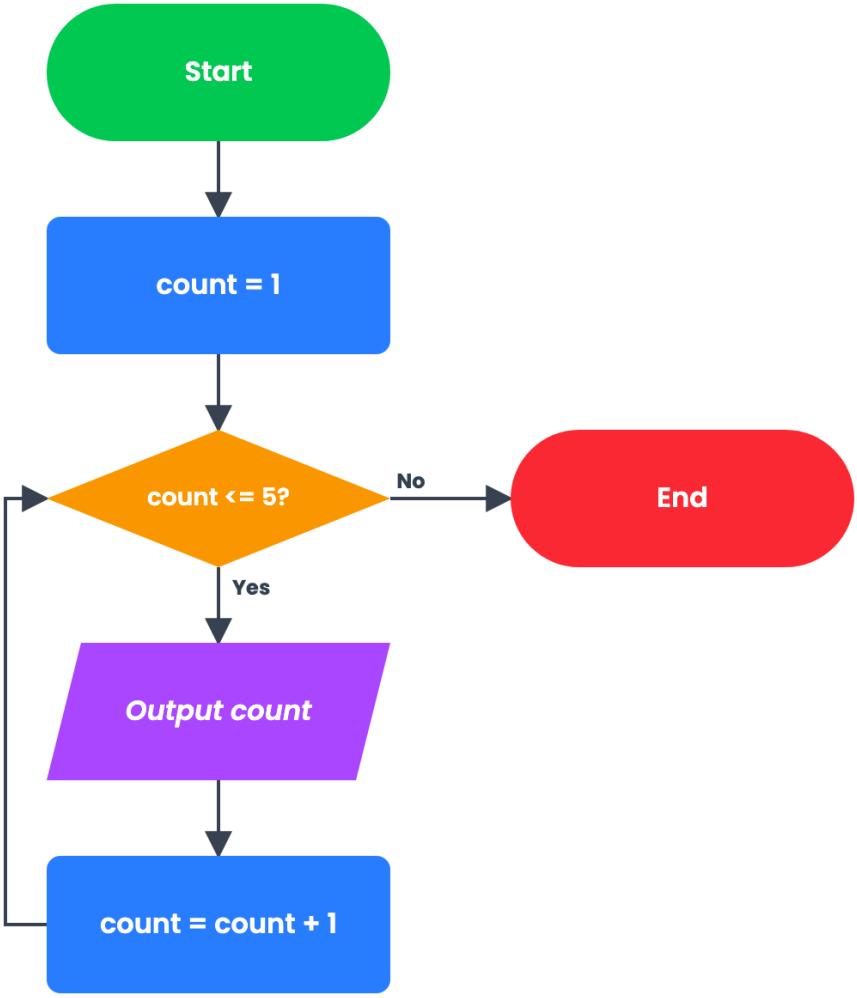


Level 12: Count to Five

Medium · Answer Key

Output the numbers 1 to 5. The arrow that goes back up is a loop. Start is already placed for you.

Answer key

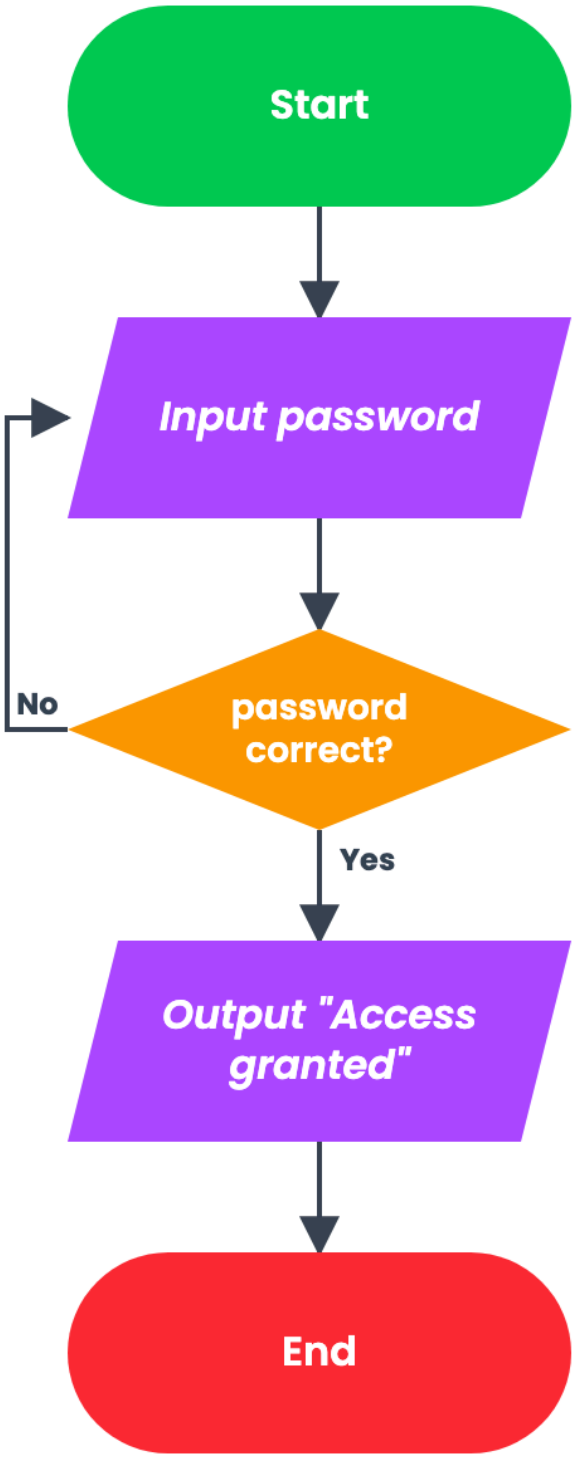


Level 13: Password Check

Medium · Answer Key

Keep asking for the password until it is correct, then grant access. Start and End are placed for you.

Answer key

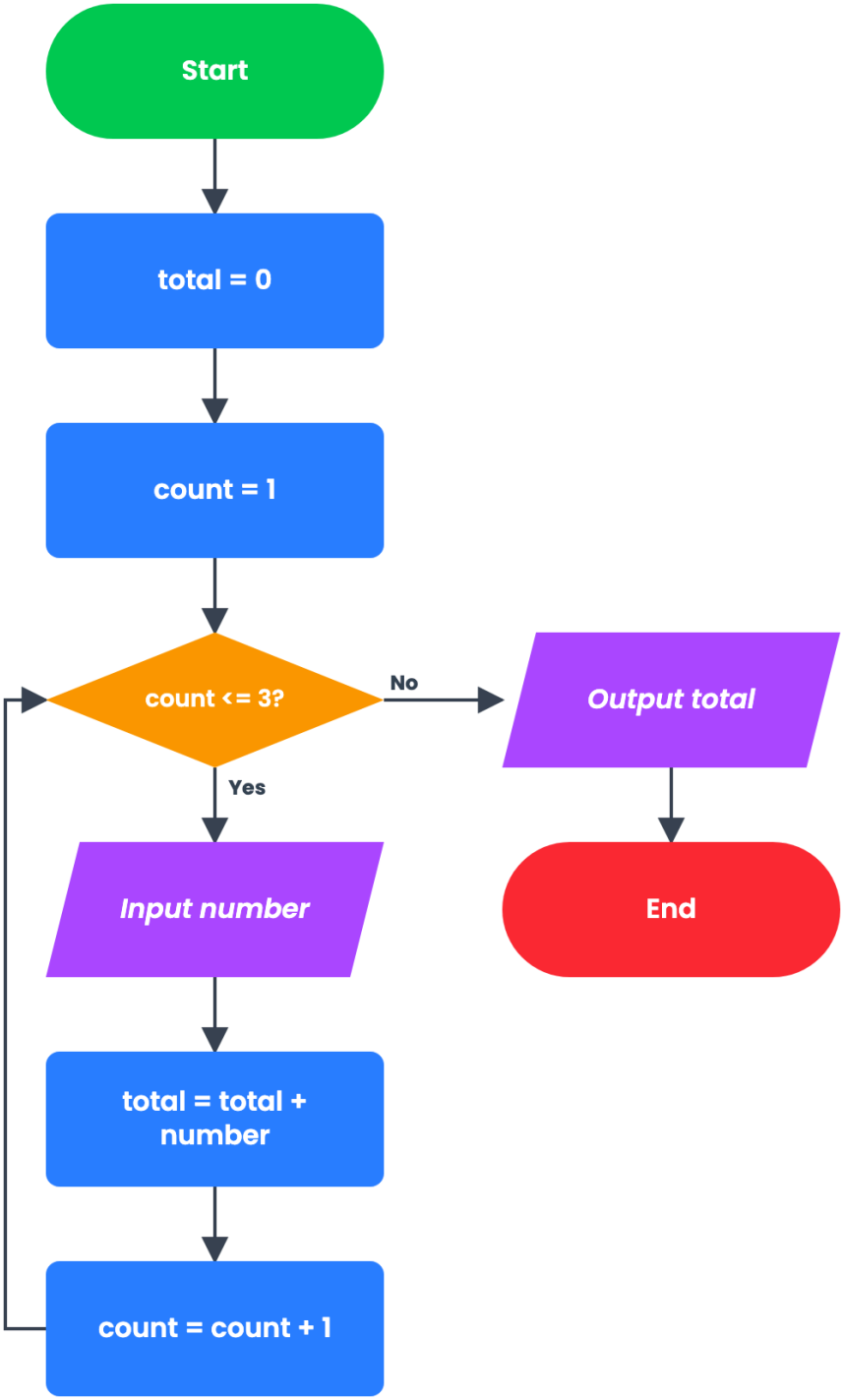


Level 14: Running Total

Medium · Answer Key

Input three numbers and output their total. This one combines a counter, a total and a loop.

Answer key

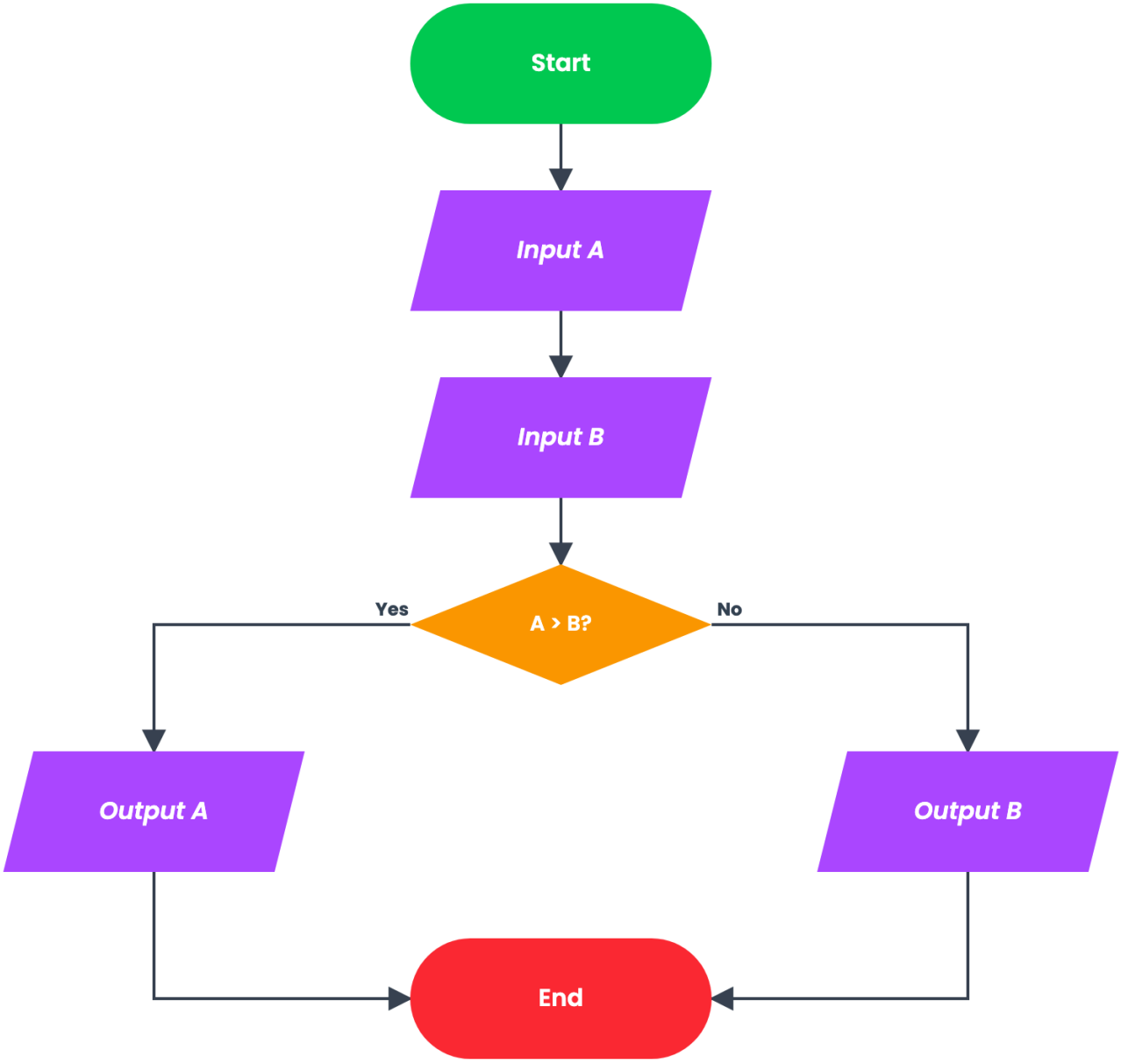


Level 15: Larger Number

Medium · Answer Key

Input two numbers and output the larger one.

Answer key

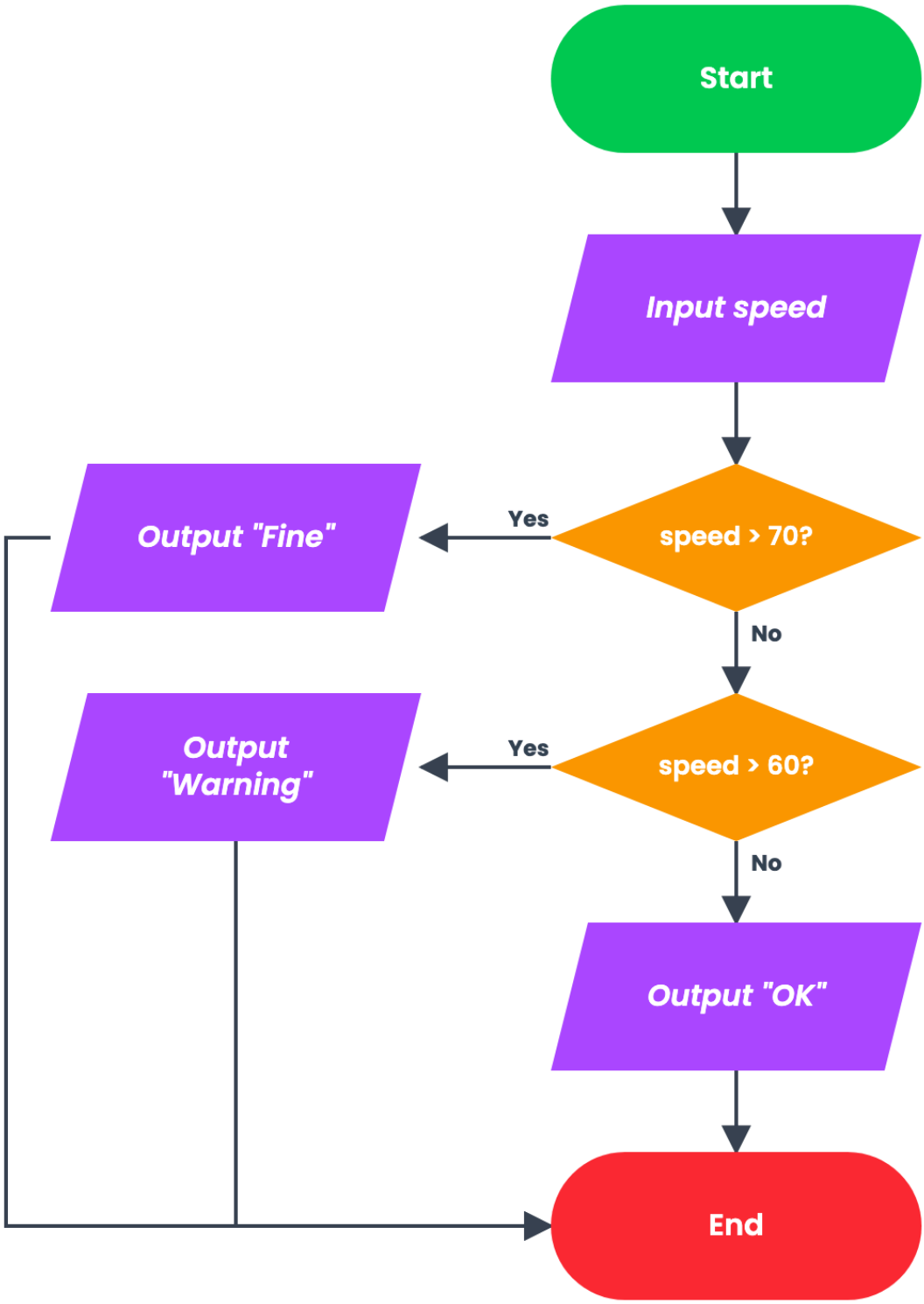


Level 16: Speed Camera

Medium · Answer Key

Over 70 is a fine. Over 60 is a warning. Anything else is fine to drive on. Start is already placed for you.

Answer key

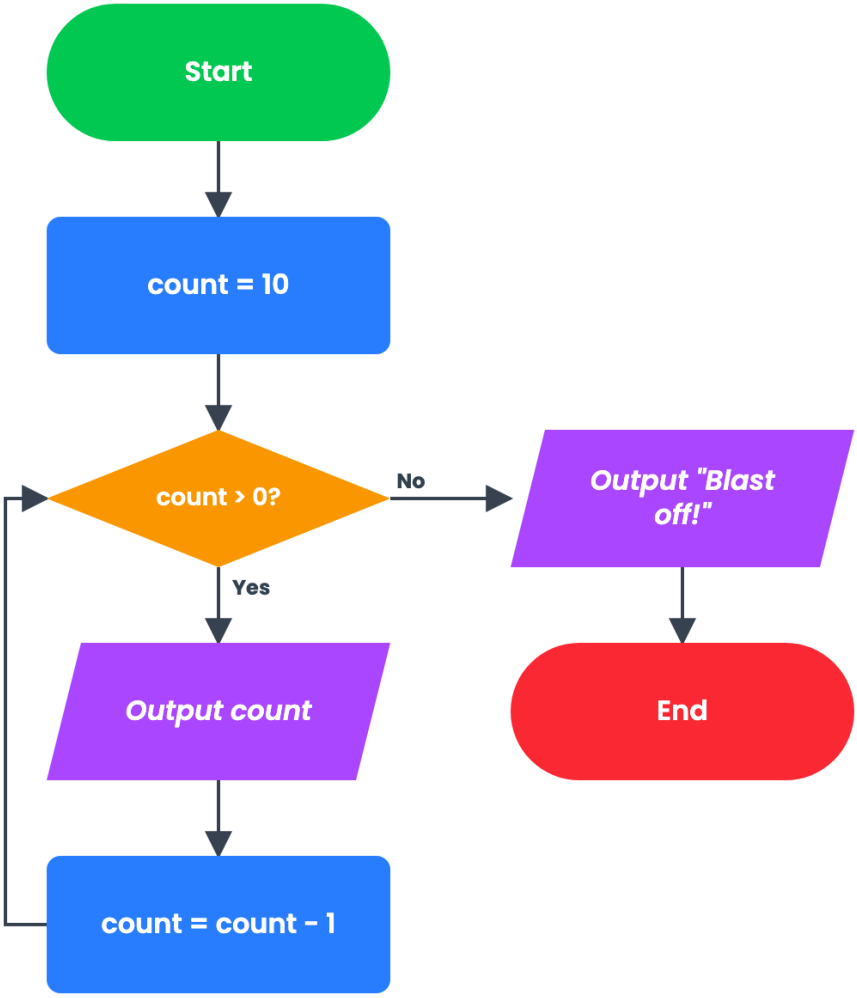


Level 17: Countdown

Medium · Answer Key

Count down from 10 and finish with a launch message. Start is already placed for you.

Answer key

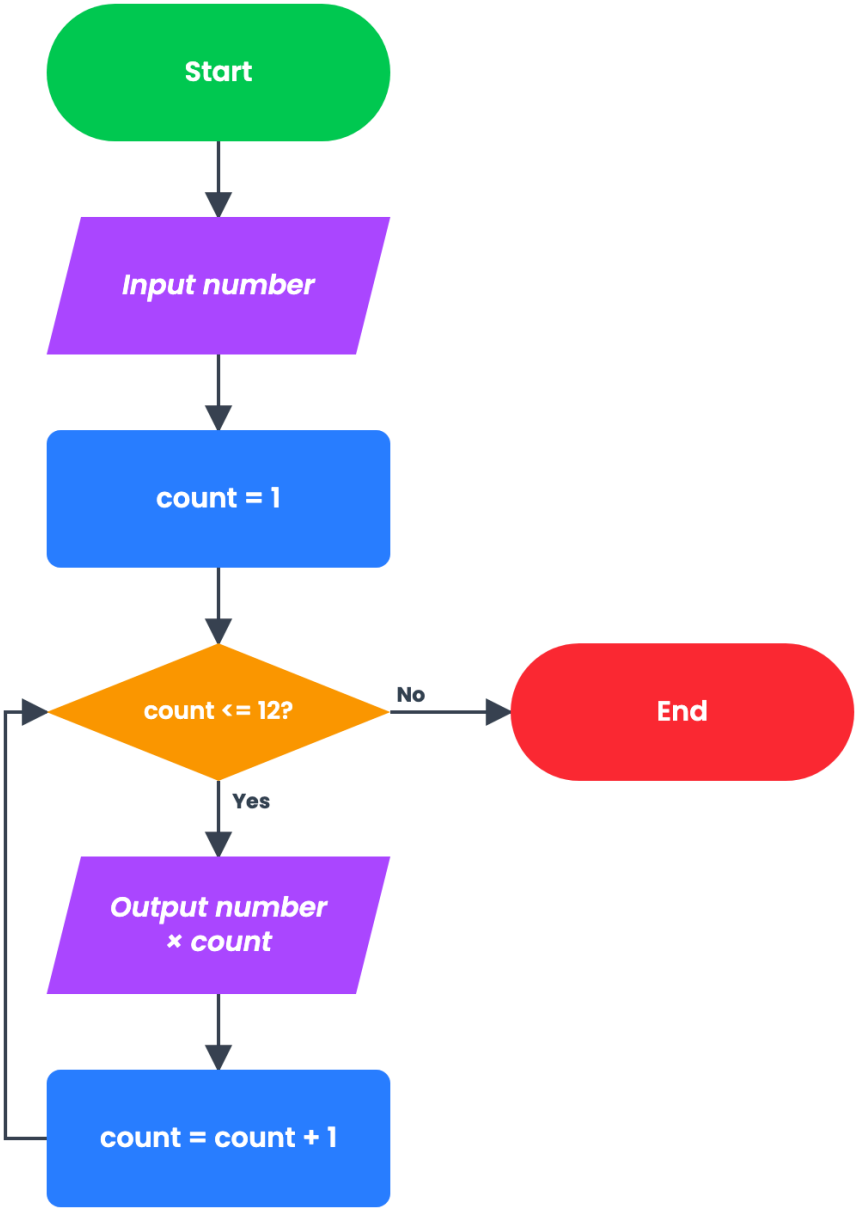


Level 18: Times Table

Medium · Answer Key

Input a number and output its 12 times table. Start is already placed for you.

Answer key

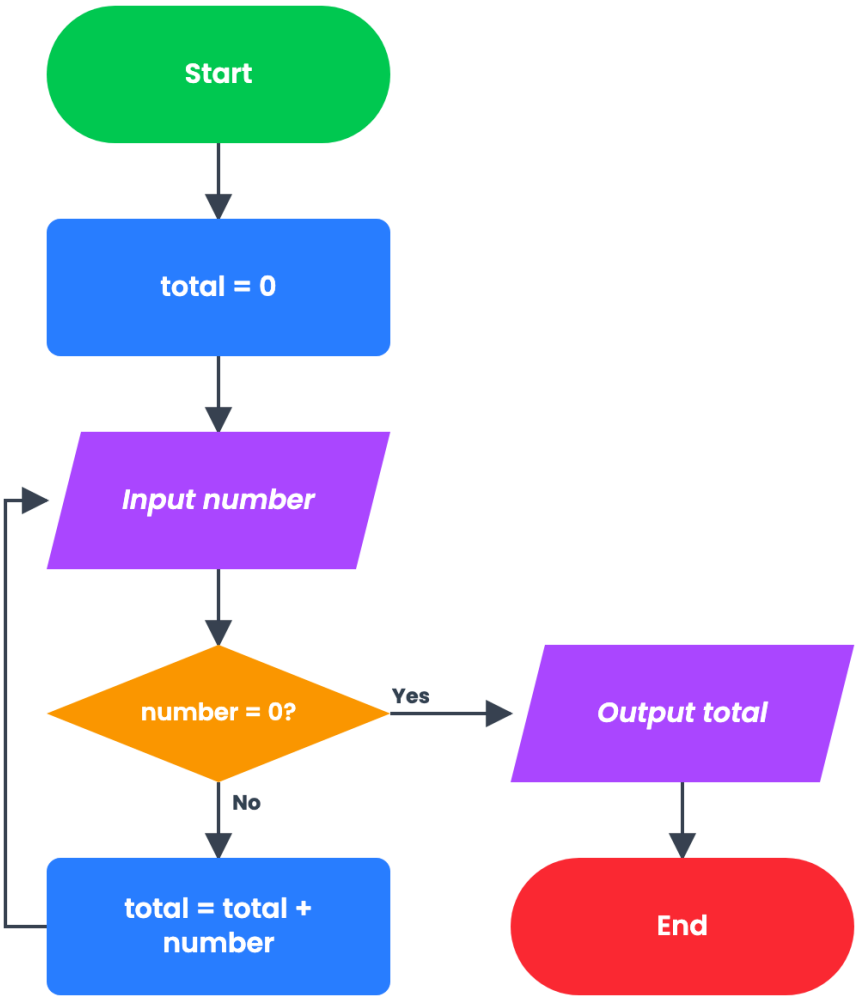


Level 19: Total Until Zero

Medium · Answer Key

Keep adding numbers to a total until the user enters 0, then output the total. Start is already placed for you.

Answer key

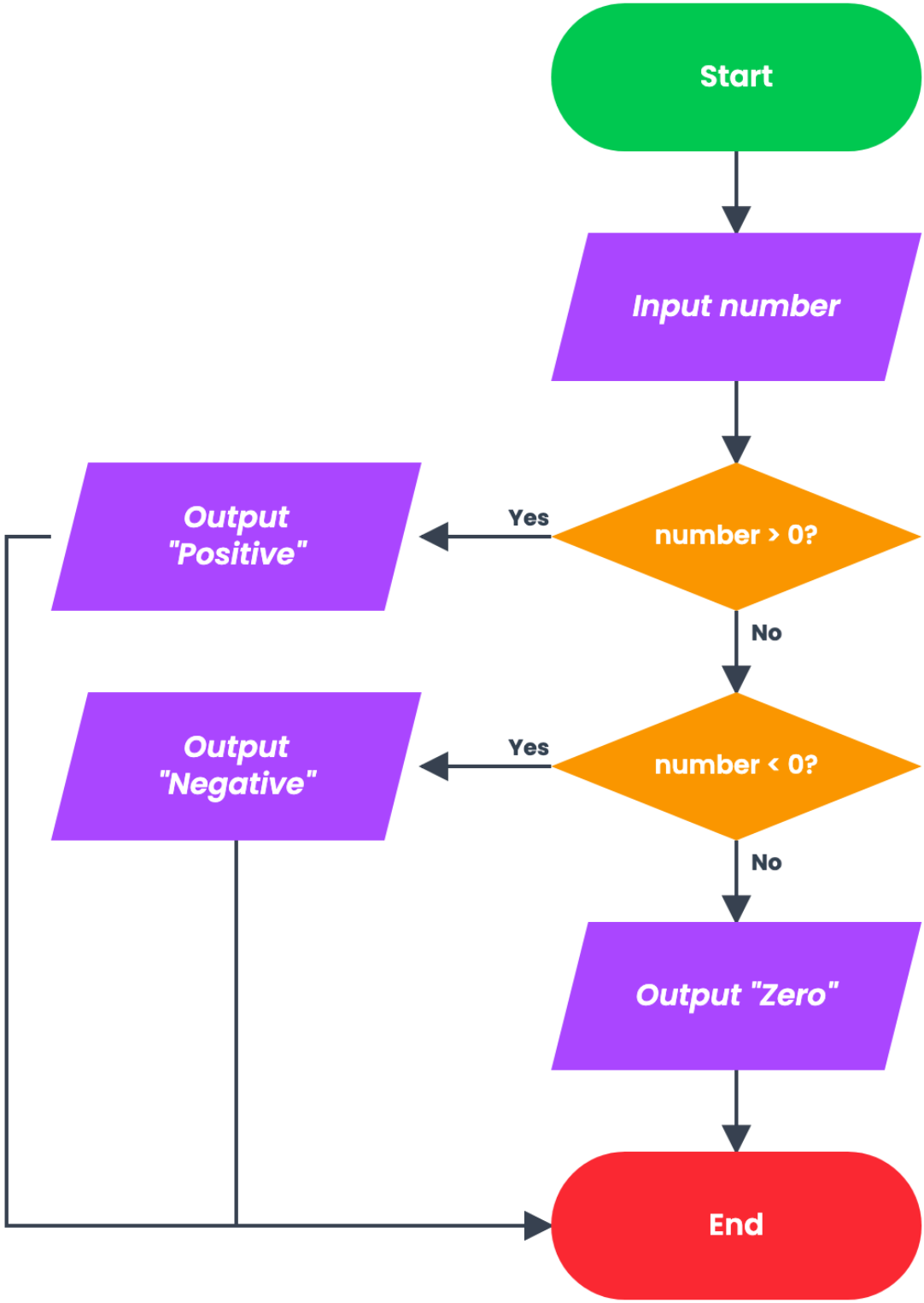


Level 20: Number Sign

Medium · Answer Key

Output whether a number is positive, negative or zero. Start is already placed for you.

Answer key

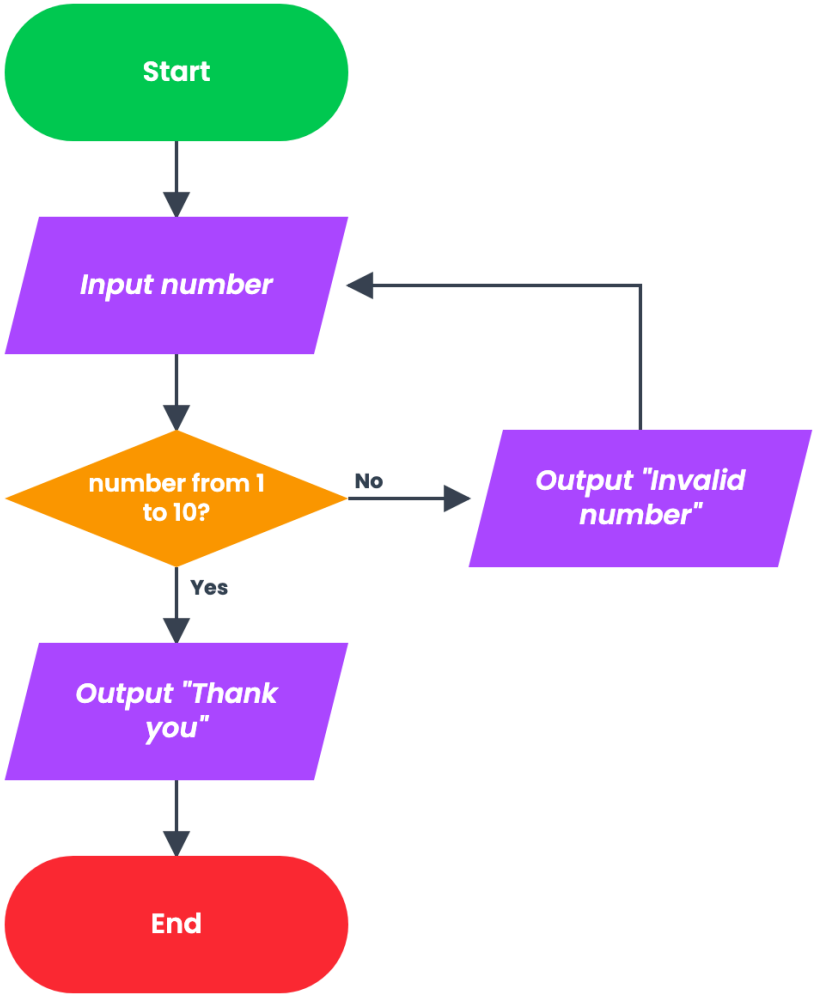


Level 21: Input Validation

Hard · Answer Key

Ask for a number from 1 to 10. If it is not in range, say so and ask again. Start is already placed for you.

Answer key

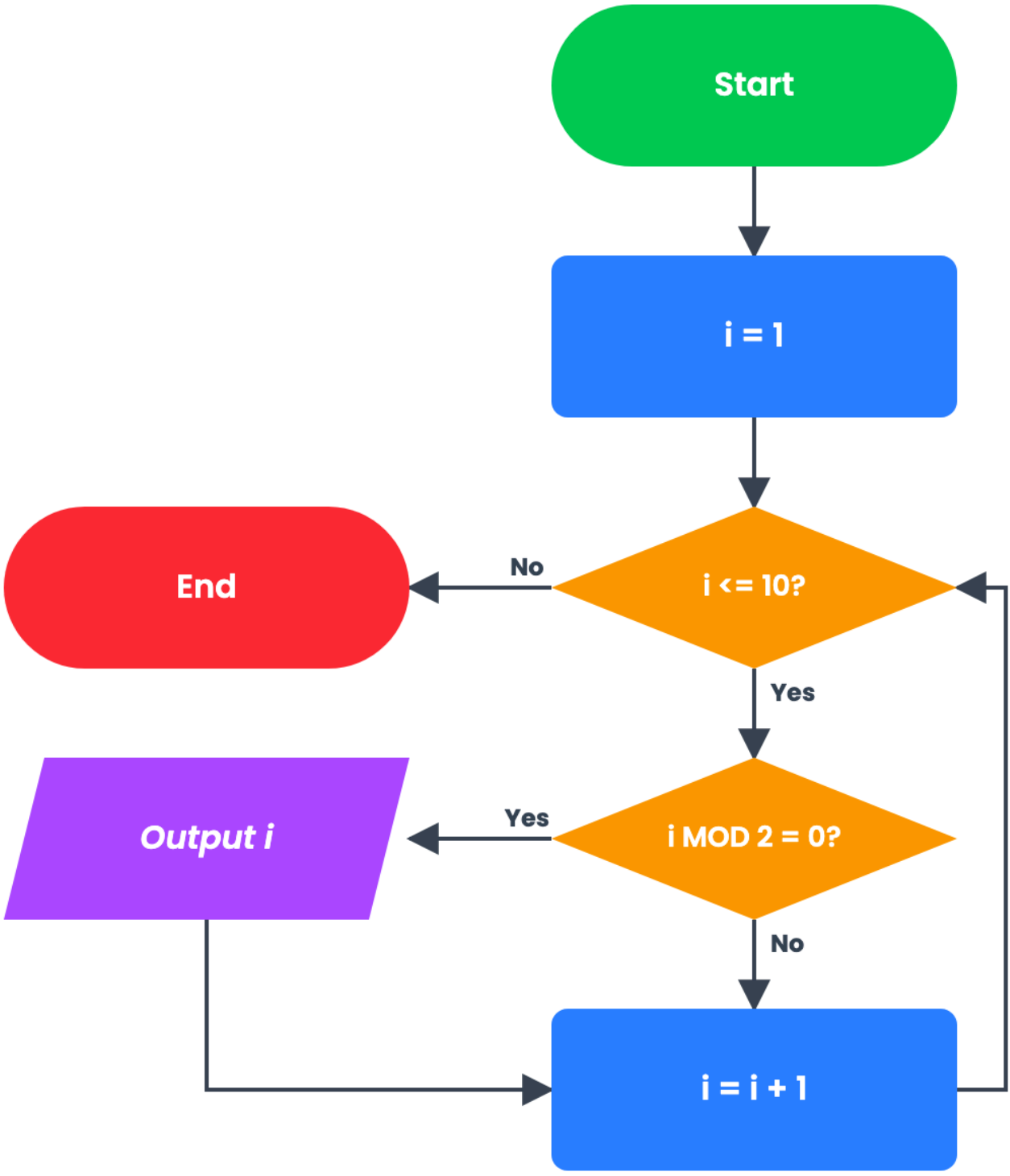


Level 22: Even Numbers

Hard · Answer Key

Output only the even numbers from 1 to 10. A decision sits inside the loop. Start is already placed for you.

Answer key

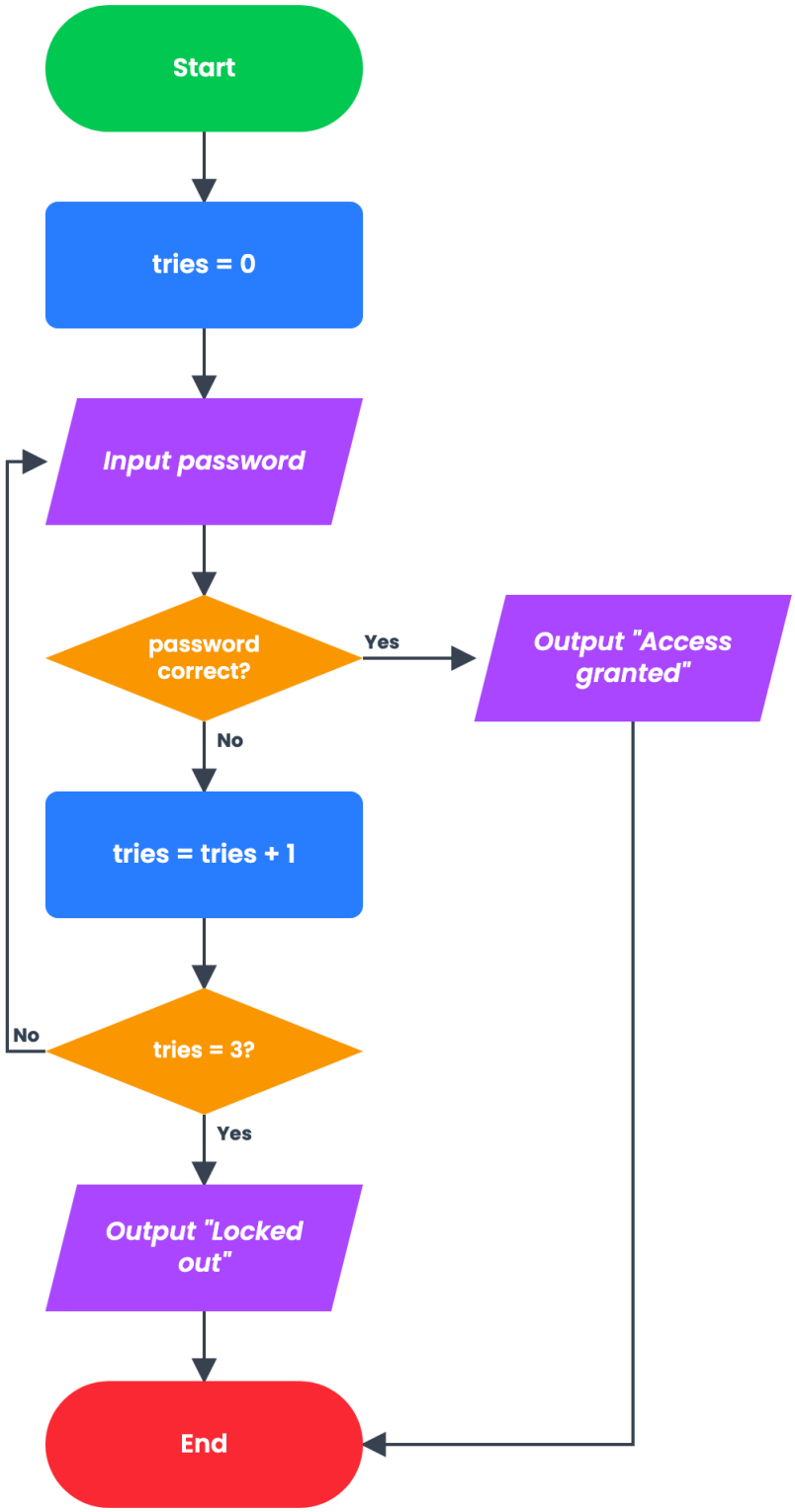


Level 23: Three Attempts

Hard · Answer Key

The user gets three tries to enter the correct password. After three wrong tries they are locked out. Start is already placed for you.

Answer key

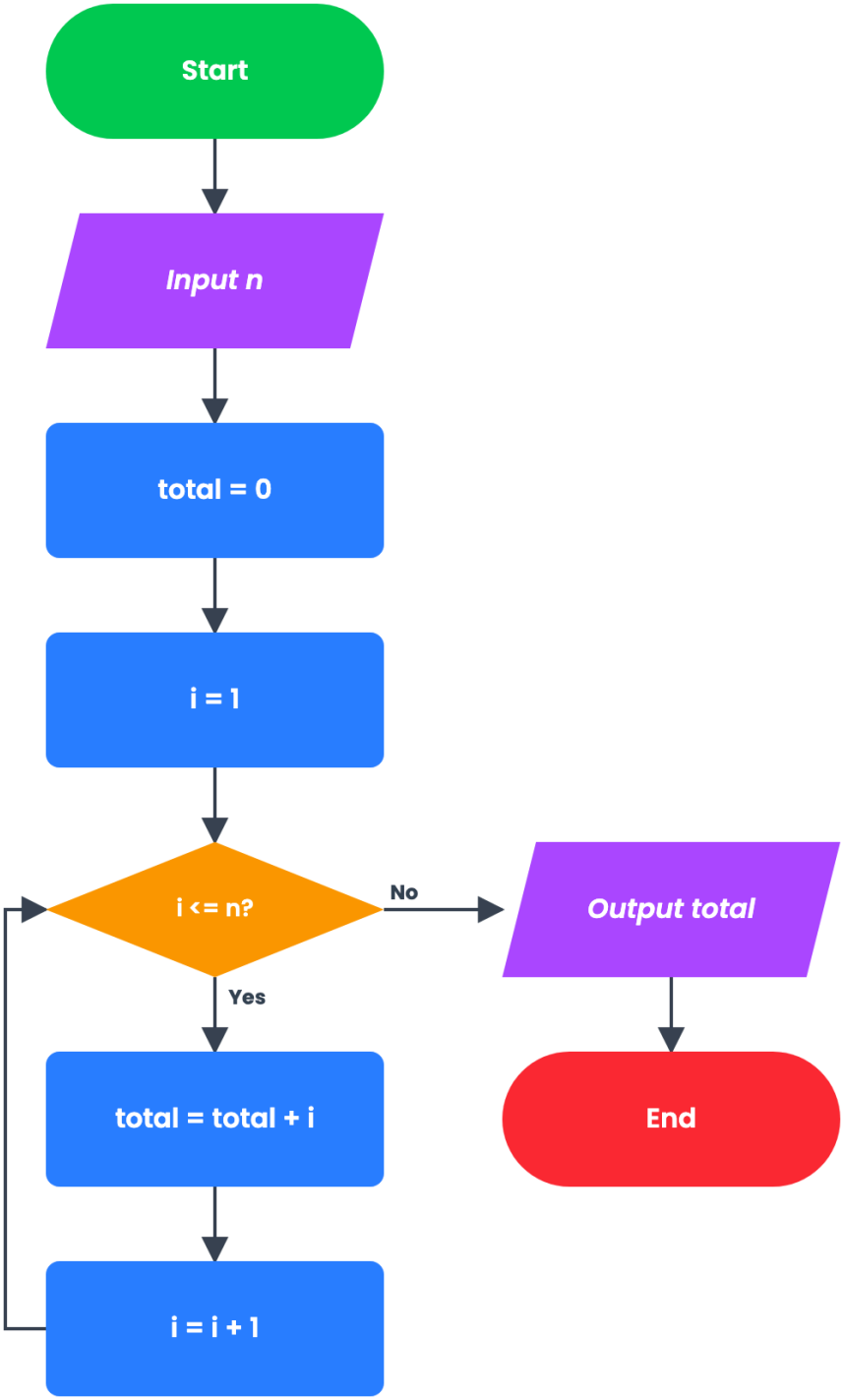


Level 24: Sum to N

Hard · Answer Key

Input a number n and output the total of 1 + 2 + … + n. Start is already placed for you.

Answer key

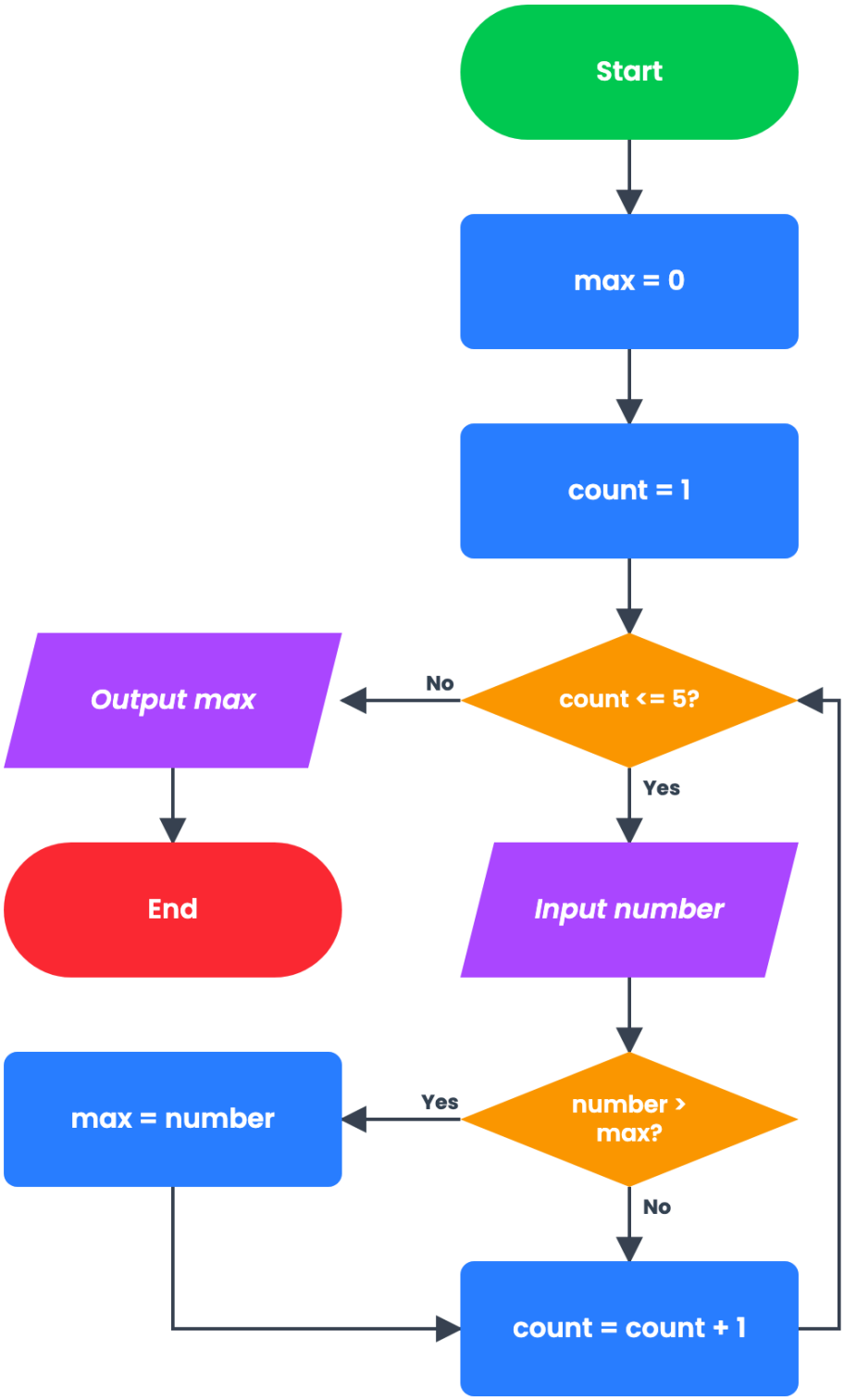


Level 25: Largest of Five

Hard · Answer Key

Input five numbers and output the largest. Keep track of the biggest so far. Start is already placed for you.

Answer key

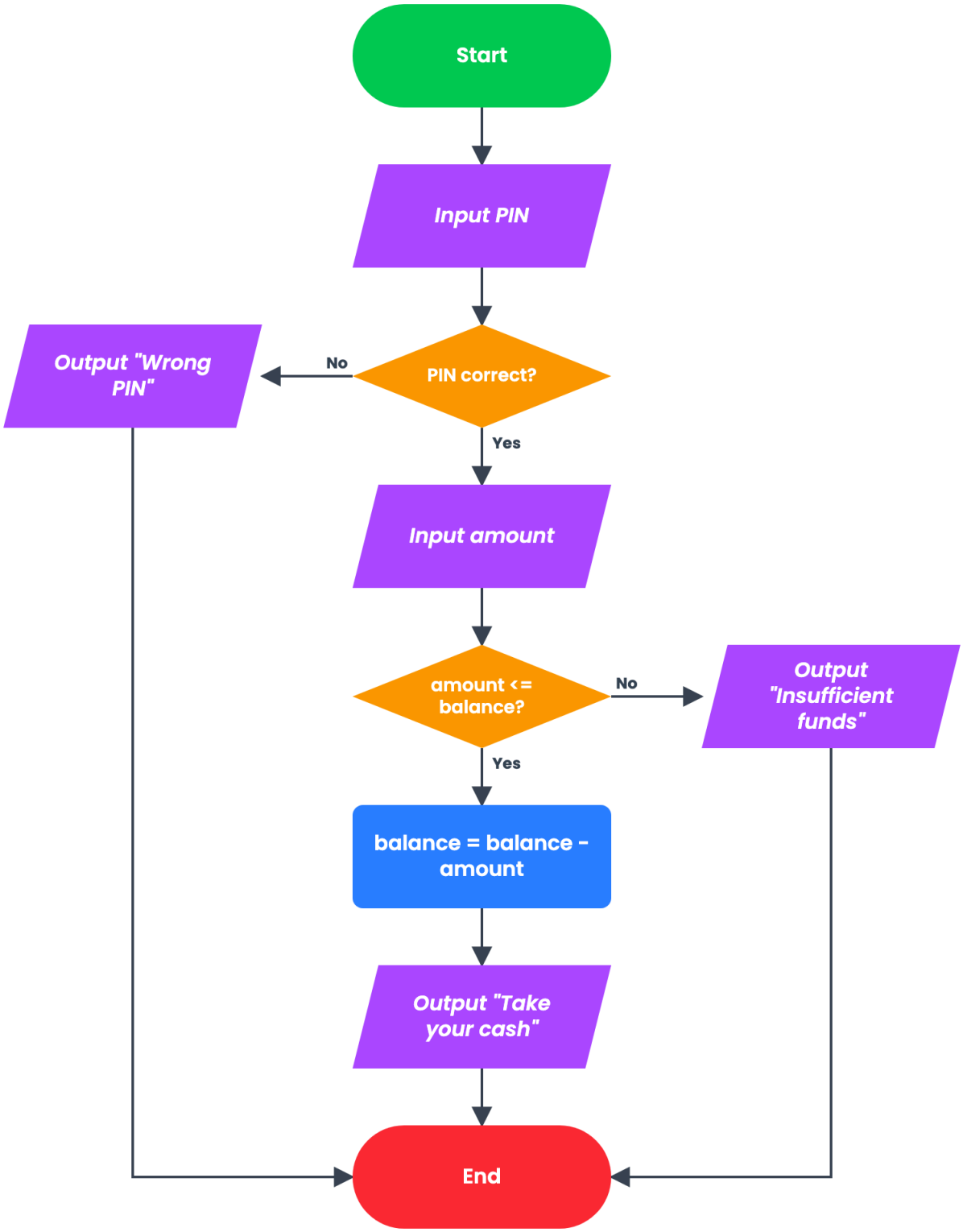


Level 26: Cash Machine

Hard · Answer Key

Check the PIN, then check there is enough money before paying out. Every path finishes at End. Start is already placed for you.

Answer key

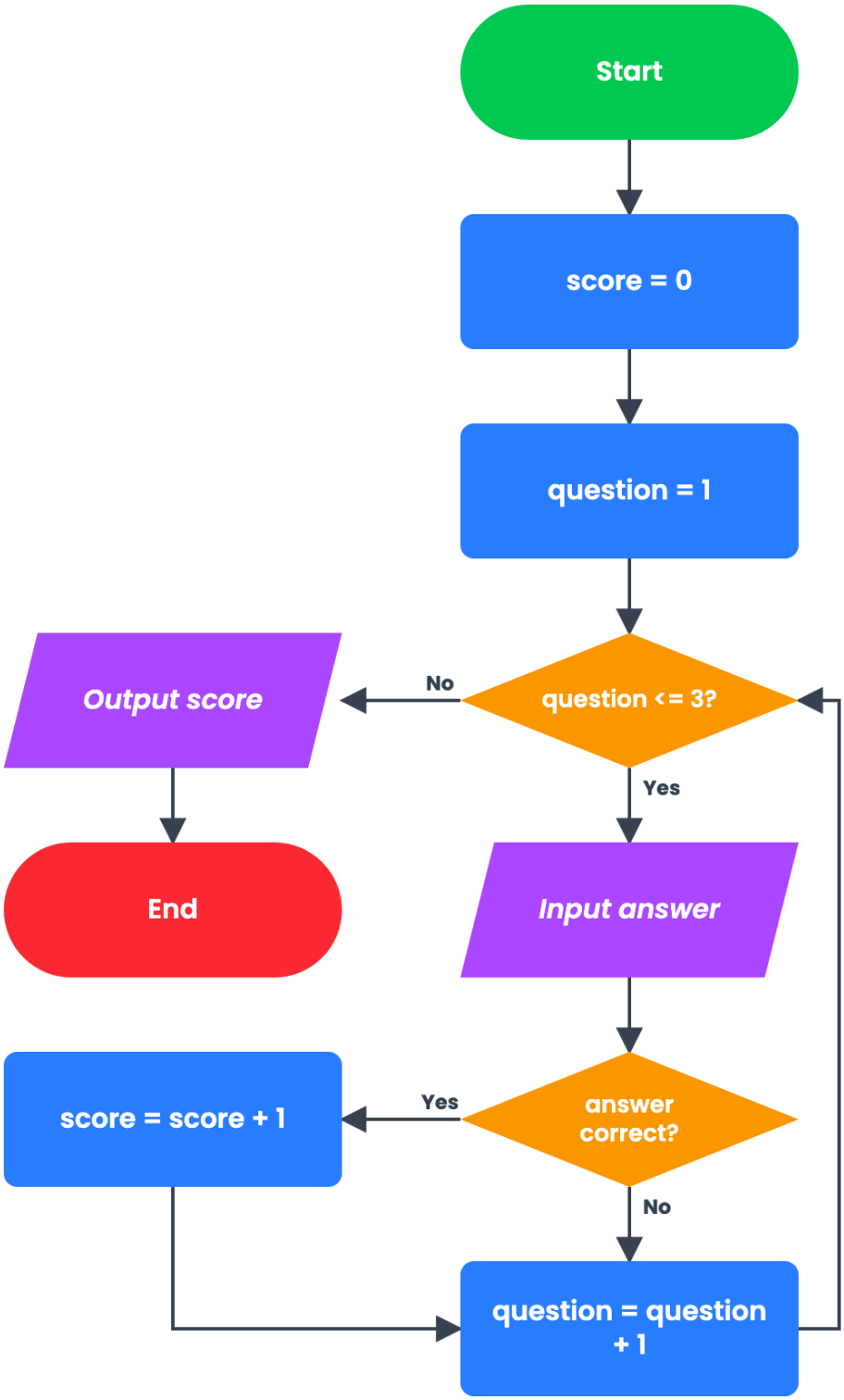


Level 27: Quiz Score

Hard · Answer Key

Ask three questions. Add one to the score for each correct answer, then output the score. Start is already placed for you.

Answer key

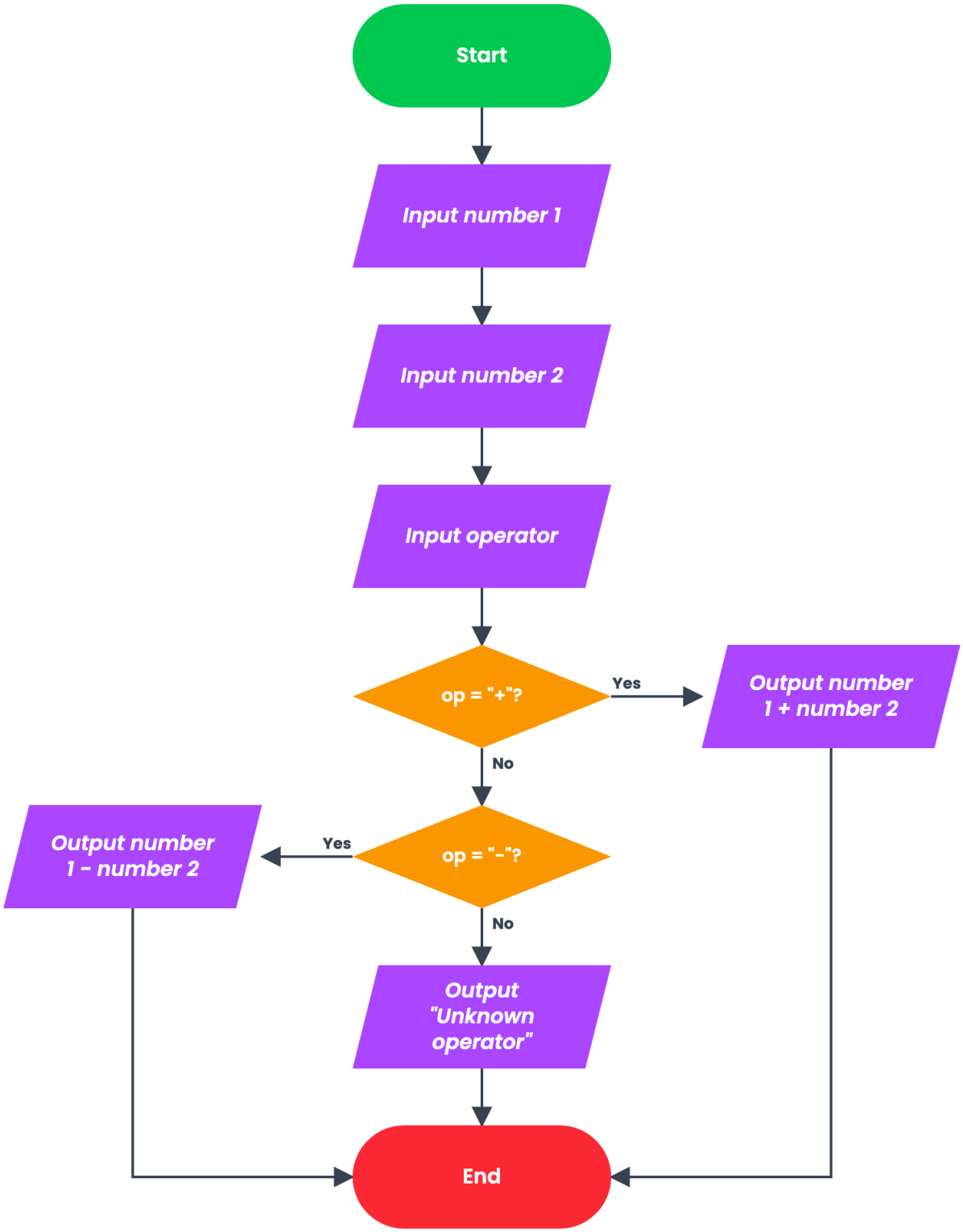


Level 28: Calculator

Hard · Answer Key

Input two numbers and an operator (+ or -), then output the answer. Anything else is an unknown operator. Start is already placed for you.

Answer key

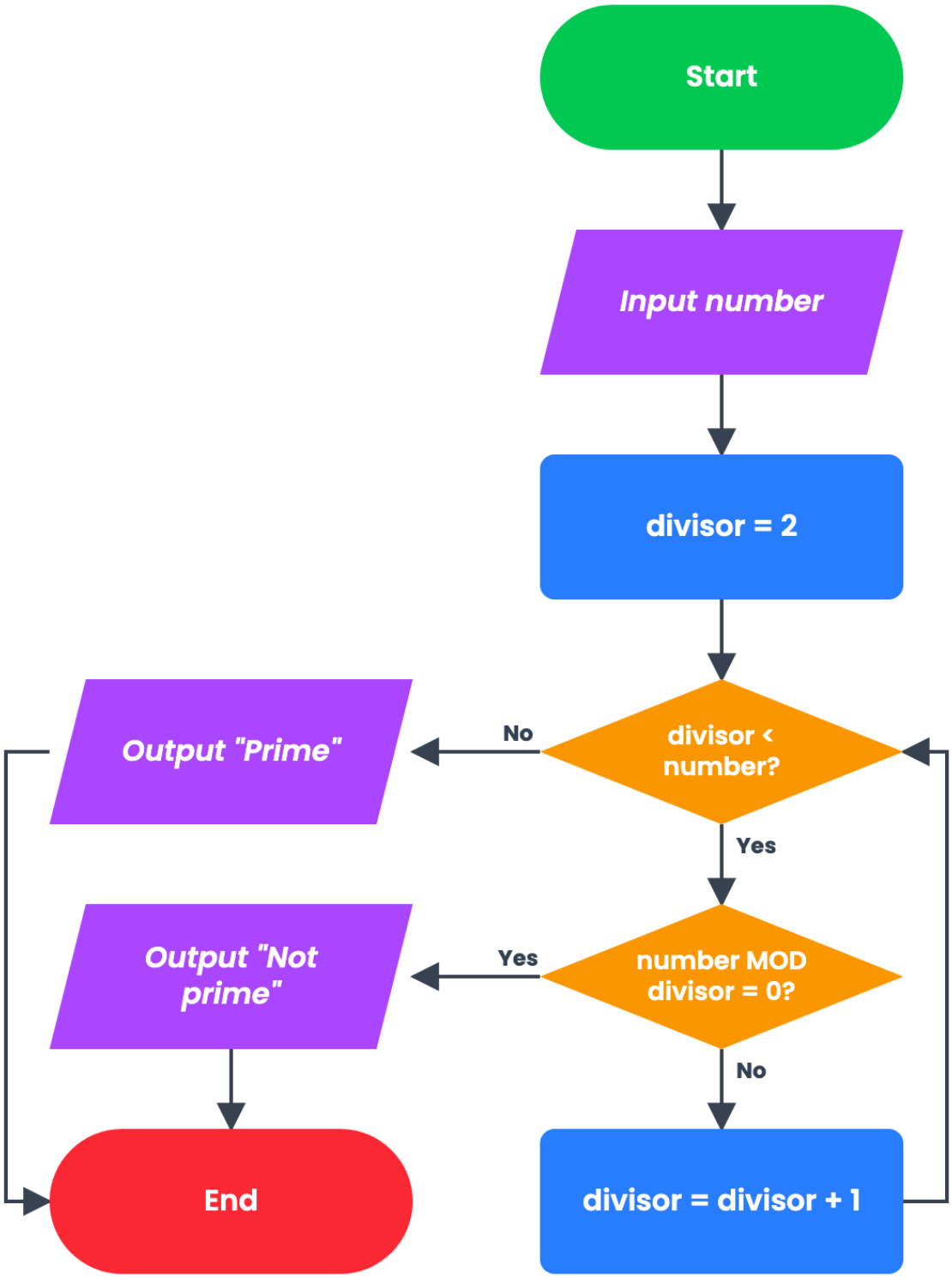


Level 29: Prime Number

Hard · Answer Key

Try dividing by 2, 3, 4... up to one less than the number. If any divide exactly it is not prime. Start is already placed for you.

Answer key



Level 30: Menu Loop

Hard · Answer Key

Show a menu: 1 says hello, 2 says goodbye, 3 quits. Any other choice asks again. Start is already placed for you.

Answer key

